NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-6106/9C(USAF)
19 December 1989

SUPERSEDING
MIL-R-6106/9B(USAF)
17 MARCH 1986

PERFORMANCE SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, CONTINUOUS DUTY, 60 AMP, 3PDT, ENVIRONMENTALLY SEALED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

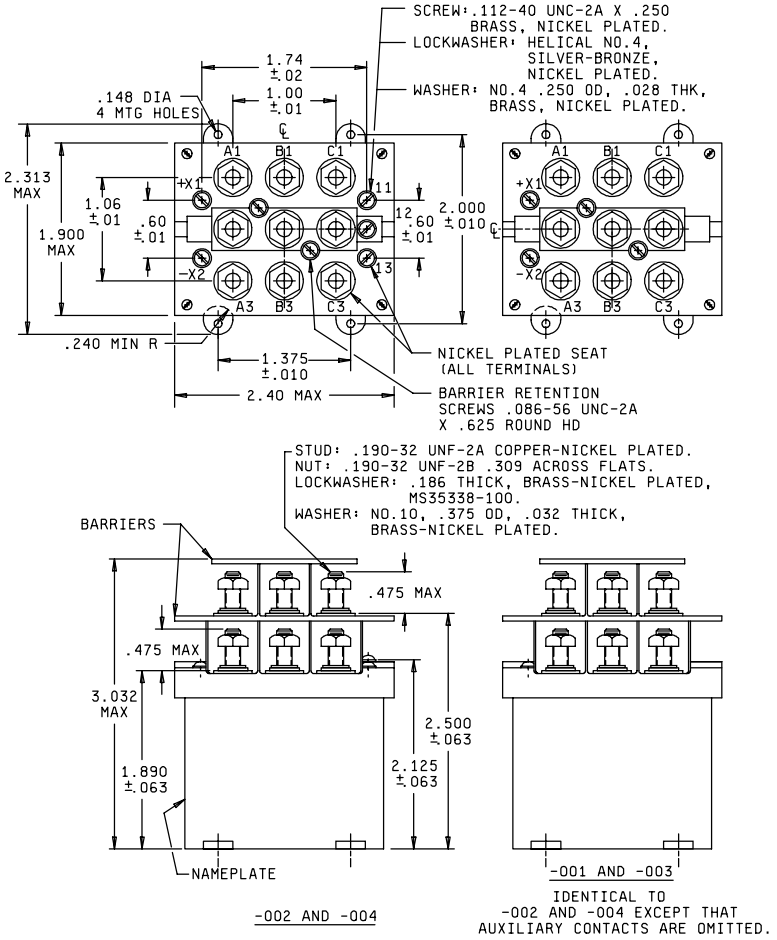
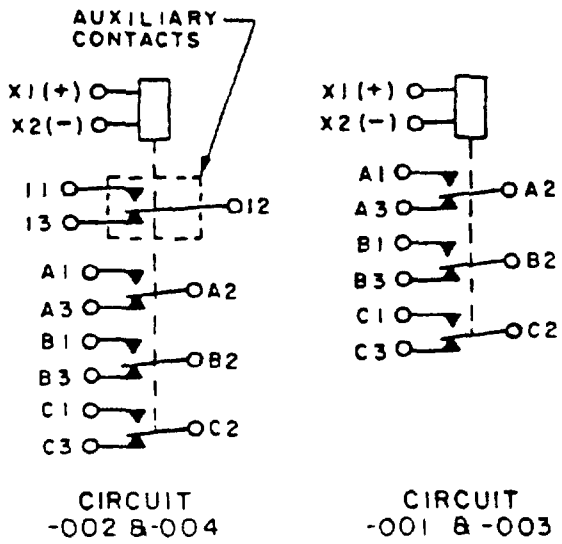


FIGURE 1. Relay, outline drawing.

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INCHES	mm	INCHES	mm	INCHES	mm
.010	0 25	.309	7.86	1.890	48.01
.02	0.51	.375	9 53	1.900	48.26
.028	0.71	.440	11.18	2.000	50.80
.032	0.81	.600	15 24	2.115	53.72
063	1.60	.625	15 88	2.313	58 75
.148	3 76	1 000	25 40	2.40	60.96
.186	4.72	1 060	26.92	2.500	63.50
.240	6 10	1.375	35 18	3.032	77 01
.250	6 35	1.74	44.20		

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Weights include cover, barriers, and terminal hardware.
5. When using relay at rated 60 amps, the use of lugs per Burndy part number YAEV6C-68 or equal is recommended
6. Terminal temperature rise under continuous current conditions is 95°C maximum

FIGURE 1 Relay, outline drawing - Continued

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TABLE I. Dash number and operating characteristics.

Dash no	Type	Coil data: Coil	Nominal: Volts	Nominal: Freq	Nominal: Res	Maximum: Volts	Maximum: Ampere: Inrush	Maximum: Ampere: Steady state	Max pickup voltage 1/: Normal 1/	Max pickup voltage 1/: High Temp Test	Max pickup voltage 1/: Cont Current Test	Hold voltage 1/	Drop out voltage 1/	Time milliseconds - maximum: Operate 2/	Time milliseconds - maximum: Release 3/	Time milliseconds - maximum: Bounce: Main: NO	Main: NC	Auxiliary: NO	Auxiliary: NC
-001	I	X1, X2	28	DC	-	29	1 25	25	17	20	21 5	7 0	1 5	25	25	2	2	-	-
002	I	X1, X2	28	DC	-	29	1 25	25	17	20	21 5	7 0	1 5	25	25	2	2	2	2
-003	I	X1, X2	115	400	-	124	4	1	95	95	100	40	10	25	35	2	2	-	-
004	I	X1, X2	115	400	-	124	4	1	85	95	100	40	10	25	35	2	2	2	2

1/ Over the temperature range 2/ With nominal coil voltage 3/ From nominal coil voltage

TABLE II Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc 1/: Main: NO	28 V dc 1/: Main: NC	28 V dc 1/: Auxiliary: NO	28 V dc 1/: Auxiliary: NC	115 V ac, 1 phase 1/: Main: 400 Hz	115 V ac, 1 phase 1/: Main: 60 Hz	115 V ac, 1 phase 1/: Auxiliary: 400 Hz	115 V ac, 1 phase 1/: Auxiliary: 60 Hz	115/200 V ac, 3 phase 1/: Main: 400 Hz	115/200 V ac, 3 phase 1/: Main: 60 Hz	115/200 V ac, 3 phase 1/: Auxiliary: 400 Hz	115/200 V ac, 3 phase 1/: Auxiliary: 60 Hz
Resistive	100	-	-	3	3	60	-	3	-	60	-	-	-
Inductive	100	-	-	1 5	1 5	60	-	1 5	-	60	-	-	-
Motor	100	-	-	-	-	40	-	-	-	40	-	-	-
Lamp		-	-	5	.5	20	-	5	-	20	-	-	-
Transfer load 2/													
Mechanical life reduced current	200	-	-	75	75	15	-	.75	-	15	-	-	-
Minimum 3/ current	50	-	-	Per spec		4	-	-	-	4	-	-	-

1/ Absence of value indicates relay is not rated for that application

2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating.

3/ Minimum current to be conducted at +71°C. Number of minimum current cycles following continuous current test - 5000.

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REQUIREMENTS

Weight 001 and 002 - 10.5 ounces maximum
003 and 004 - 11.5 ounces maximum.

Temperature range -65°C to +71°C.

Maximum altitude rating: 50,000 feet 1/.

Shock g-level 1/2 sine, 25 g's.

Duration 6 to 9 ms.

Maximum duration contact opening. 1 ms.

Vibration-sinusoidal: See table III.

TABLE III. Vibration-sinusoidal.

Dash No.	Vibration level					
	5-10 Hz	10-55 Hz	55-400 Hz	400-800 Hz	800 - 2000 Hz	
					-55°C and 25°C	+71°C
-001	.08 DA	.06 DA	10 g's	10 g's	10 g's	7 g's
-002			10 g's	8 g's	8 g's	7 g's
-003			10 g's	10 g's	10 g's	7 g's
-004			10 g's	8 g's	8 g's	7 g's

Vibration (random) Not applicable.

Acceleration 15 g's.

Seal:

Applicable specification: MIL-STD-202, method 112, test condition C, procedure IV

Leak rate 6×10^{-4} STD cc/s 2/.

Terminal strength

Applicable specification MIL-STD-202, method 211, test condition E.

Torque 24 pound-inches.

Insulation resistance, initial 200 megohms minimum.

After life or enviromental tests: 100 megohms minimum.

1/ Application range can be extended to 80,000 feet by suitably encapsulating the terminals

2/ During the test, pressure shall be reduced below 10 microns for a minimum of 10 seconds.

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Dielectric withstanding voltage (sea level): 2 to 5 seconds.

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1250	1500	1000	1125
Auxiliary contacts	1250	1250	1000	1000
Across open power contacts	---	1250	---	625
All other points	1800	1800	1350	1350

Dielectric withstanding voltage (altitude): 1 minute.

	28 V dc	115 V ac
Coil to case	500	500
Auxiliary contacts	500	500
All other points	700	700

Maximum contact drop, initial .150 volt.

After life test .200 volt.

	NO	NC
Overload current 3/	320	320
Rupture current	400	400

Duty rating: Continuous.

RFI specification (applicable to coil circuits of ac operated relays): MIL-STD-461 and MIL-STD-462.

Part number M6106/9- (dash number from table I).

TABLE IV. Qualification by similarity.

Dash number	Loads						Dynamics 1/			Environmental 2/		
	Type											
	A	B	C	D	E	F	A	X	Y	A	B	C
-001	4						2			4		
-002	4 2/						2			4		
-003		1						1			1	
-004		1 2/						1			1	

1/ All units must be tested, reference the appendix of MIL-R-6106, construction, internal.

2/ Test unit with auxiliary contacts, reference the appendix of MIL-R-6106, construction, internal.

3/ Number of rupture cycles after minimum current test - 5.

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Custodian
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent
DLA - ES

(Project 5945-F703-01)

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